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Quiz: Embeddings & Vector Databases

Section A: Multiple Choice Questions (MCQs)

1. What is an embedding in AI?
a) A database table
b) A numerical representation of text, images, or data
c) A programming language
d) A type of neural network

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2. Why are embeddings useful in AI applications?
a) They compress files
b) They convert data into numbers that machines can compare
c) They increase internet speed
d) They replace databases

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3. Which technique is commonly used to measure similarity between embeddings?
a) Binary search
b) Cosine similarity
c) Linear regression
d) Sorting algorithm

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4. What is the main purpose of a vector database?
a) Store only text documents
b) Store and search numerical vector representations efficiently
c) Store images only
d) Replace relational databases completely

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5. Which of the following is a popular vector database?
a) MySQL
b) Pinecone
c) Excel
d) Oracle Forms

Section B: True / False

6. Embeddings convert text or images into numerical vectors.
True / False

7. Vector databases are optimized for similarity search.
True / False

8. Cosine similarity measures how similar two vectors are in direction.
True / False

9. Traditional SQL databases are always faster than vector databases for similarity search.
True / False

Section C: Short Answer

10. In simple words, what is similarity search?

ANSWER:- Similarity search is a technique used to find items that are most similar to a given query based on vector embeddings instead of exact keyword matching.

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11. Give one real-world application of embeddings.

Examples may include search engines, recommendation systems, or chatbots.

ANSWER:- Example: **Recommendation systems** (e.g., suggesting products, movies, or courses based on similarity).

Other examples:

- Search engines

- Chatbots
- Image recognition

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12. What happens when two embeddings are very close in vector space?

ANSWER:- It means the two pieces of data are very similar in meaning or context.

Example:
"running shoes" and "jogging sneakers"

Section D: Practical Thinking Questions

13. Scenario:
You build a chatbot for a university website.

The chatbot should answer questions from **course PDFs and policy documents**.

Question:
Why would embeddings + a vector database be useful here?

ANSWER:- Embeddings convert **PDF content and student queries into vectors**, and the vector database quickly finds the **most similar information from documents**, allowing the chatbot to answer questions even if the wording is different.

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14. Scenario:
A user searches for **"cheap smartphones"**, but the database contains products labeled **"budget mobile phones."**

Question:
How can embeddings help retrieve the correct results even if the exact words are different?

ANSWER:- mbeddings capture **semantic meaning**, so both phrases will have **similar vectors**. The system can retrieve "budget mobile phones" even though the words are different.

Section E: Mini Hands-On Question

15. Steps Challenge

Arrange the steps for building a **semantic search system** in the correct order:

a) Store embeddings in a vector database
b) Convert documents into embeddings
c) Convert the user query into an embedding
d) Find the most similar vectors

- **ANSWER:- b → a → c → d**

b) Convert documents into embeddings
a) Store embeddings in a vector database
c) Convert the user query into an embedding
d) Find the most similar vectors